Name: Key

6. a) Give a basis for the column space of A; or explain why one does not exist.

$$A = \begin{pmatrix} 1 & 3 & 10 & 5 \\ 2 & 6 & 8 & 6 \\ 3 & 9 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 \\ 2 & 6 & 8 & 6 \\ 3 & 9 & 0 & 5 \end{pmatrix} \xrightarrow{R2-2R1} \begin{pmatrix} 1 & 3 & 10 & 5 \\ 0 & 0 & -12 & -4 \\ 3 & 9 & 0 & 5 \end{pmatrix} \xrightarrow{R3-3R1} \begin{pmatrix} 1 & 3 & 10 & 5 \\ 0 & 0 & -12 & -4 \\ 0 & 0 & -30 & -10 \end{pmatrix}$$

$$\xrightarrow{-\frac{1}{4}R2} \begin{pmatrix} 1 & 3 & 10 & 5 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & -30 & -10 \end{pmatrix} \xrightarrow{R3+10R2} \begin{pmatrix} \boxed{1} & 3 & 10 & 5 \\ 0 & 0 & \boxed{3} & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ REF.}$$

Since the first and third columns of this matrix are a basis for its column space, the same is true for the first and third columns of A.

So a basis for col(A) is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 10 \\ 8 \\ 0 \end{pmatrix} \right\}$

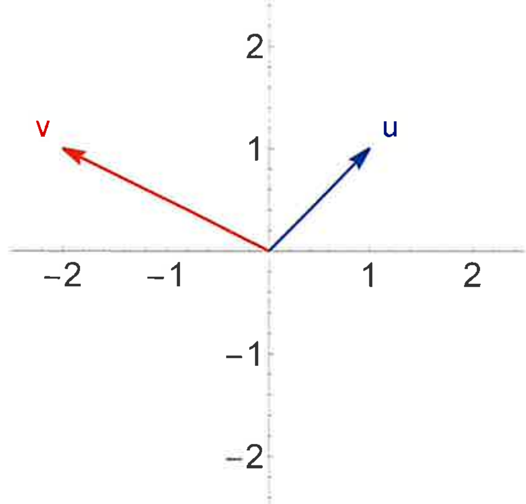
b) Give an example of a basis for $\mathbb{C}^3$ that contains the two vectors $\begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$ and $\begin{pmatrix} i-1 \\ -i-1 \\ 0 \end{pmatrix}$, or explain why this is not possible.

The vectors in a basis are linearly independent.

But $\begin{pmatrix} i-1 \\ -i-1 \\ 0 \end{pmatrix} = \underbrace{(i-1)}_{\text{scalar}} \cdot \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$. So these two vectors cannot be in a basis for $\mathbb{C}^3$ together.

Name: Key

7. a) Using the standard basis, give the matrix for the linear transformation of $\mathbb{R}^2$ that sends $\mathbf{u}$ to $\mathbf{v}$ and sends $\mathbf{v}$ to $\mathbf{u}$. The vectors are pictured below, in the standard coordinate system.



In the basis u, v this linear transformation is represented by the matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

To write this linear transformation in the standard basis, apply a change of basis:

$$\underbrace{\begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}}_{\text{convert from } u,v \text{ basis to standard basis}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \underbrace{\begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}^{-1}}_{\text{convert from standard basis to } u,v \text{ basis}} = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1/3 & 2/3 \\ -1/3 & 1/3 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1/3 & 2/3 \\ -1/3 & 1/3 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}}$$

b) Recall that a symmetric matrix is a square matrix such that $A = A^T$. Give an example of a matrix that is similar to a symmetric matrix but that is not itself symmetric; or explain why this is not possible.

For instance, we above showed that $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}$ are similar. And $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is symmetric but $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}$ is not.

Check of part (a): $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ ✓

$\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ ✓